Omkar Badadale

Data Scientist

USA | omkarbadadale09@gmail.com | +1 508-287-9861

SUMMARY

Data Scientist with 4+ years of experience developing and evaluating machine learning models and analytical solutions across financial services, telecommunications, and consulting. Skilled in feature engineering, predictive modeling, customer segmentation, churn analysis, and model explainability using Python, SQL, PySpark, and scikit-learn. Experienced in NLP and prototyping LLM-powered retrieval applications, with a focus on model reliability, data quality, and translating analytical findings into actionable business recommendations.

SKILLS

Programming & Data Analysis: Python, SQL, Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Visualization
Statistics & Machine Learning: Statistical Analysis, Hypothesis Testing, Feature Engineering, Logistic Regression, Random Forest, XGBoost, K-Means Clustering, Classification, Anomaly Detection, Cross-Validation, Hyperparameter Tuning
Model Evaluation & Explainability: ROC-AUC, Precision, Recall, F1-Score, Model Validation, Threshold Optimization, SHAP, Model Monitoring, Responsible AI
NLP & Generative AI: Natural Language Processing (NLP), TF-IDF, Sentiment Analysis, Topic Modeling, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Text Embeddings, Vector Search, LLM Evaluation
Big Data, Cloud & MLOps: PySpark, Apache Spark, Spark SQL, AWS S3, Amazon EMR, Amazon SageMaker, MLflow, Experiment Tracking, Git
Business Intelligence: Power BI, KPI Reporting, Customer Analytics, Churn Analysis, Financial Risk Analytics, Stakeholder Communication

EXPERIENCE

Citi, USA | Data Scientist — Mar 2026 - Current

- Engineered customer- and transaction-level features from banking records using Python, SQL, and PySpark to prepare analytical datasets for financial risk assessment and suspicious-activity modeling.
- Developed XGBoost classification models and anomaly-detection workflows to prioritize potentially suspicious transactions, achieving a ROC-AUC of 0.88 on a held-out test dataset.
- Applied SHAP explainability to identify influential transaction features and translate model predictions into interpretable risk drivers for analyst review and investigation prioritization.
- Evaluated classification thresholds and false-positive trade-offs on validation data, identifying a threshold associated with a 15% reduction in false-positive alerts while monitoring detection performance.
- Prototyped a retrieval-augmented generation (RAG) application using an LLM, text embeddings, and vector search across 2,000+ approved compliance documents to retrieve relevant policy passages for analyst research.
- Evaluated the RAG prototype against a 200-question test set covering retrieval relevance, answer consistency, and citation accuracy; documented quality limitations and access-control requirements for potential deployment.

VOIS (Vodafone Intelligent Solutions), India | Data Scientist — Aug 2022 - Aug 2024

- Engineered subscriber-level analytical datasets from 1M+ usage, recharge, billing, and customer-care records using SQL and PySpark, creating behavioral features for telecom churn modeling.
- Developed and evaluated XGBoost and Random Forest churn models, achieving a ROC-AUC of 0.84 on a held-out validation dataset to identify subscribers at elevated risk of leaving.
- Analyzed recharge gaps, declining data usage, repeated complaints, and service interruptions to identify behavioral churn indicators and characterize attrition patterns across subscriber segments.
- Applied NLP-based sentiment classification to 40K+ customer complaints and service comments, categorizing recurring network, billing, and service-quality issues to inform customer experience analysis and retention planning.
- Integrated churn predictions with subscriber tenure, plan characteristics, and customer segmentation to prioritize high-risk subscribers and inform targeted retention strategies.
- Optimized recurring data-preparation workflows using PySpark and Spark SQL, reducing processing time by 25% on the measured workload and improving the efficiency of telecom analytics.

Fractal Analytics, India | Junior Data Scientist — Jan 2021 - Jul 2022

- Analyzed 250K+ customer transaction and demographic records using Python, SQL, and Pandas to develop customer profiles and purchasing insights for client marketing strategies.
- Applied K-means clustering and statistical profiling to segment customers by purchase frequency, average spend, and recency, helping marketing teams tailor campaigns to distinct audience groups.
- Built Logistic Regression and Random Forest propensity models to estimate campaign response likelihood, improving targeting precision by 12% against the established baseline during model evaluation.

- Applied TF-IDF and topic modeling to 15K+ customer feedback comments, identifying recurring product concerns and service themes for client reporting and customer experience recommendations.
- Automated recurring data-cleaning and exploratory analysis tasks with reusable Python workflows, saving approximately 6 hours per week in manual preparation across analytical deliverables.
- Developed Power BI dashboards and stakeholder reports to communicate customer trends, segment profiles, and model findings, supporting data-informed marketing and business decisions.

PROJECTS

Graduate Research

- Developed supervised machine learning and statistical models using ESG and financial data covering 1.35K companies across 33 countries, applying regression, hypothesis testing, and exploratory analysis to explain approximately 50% of the variation in ESG pillar weights.
- Engineered predictive features and reusable Python model-evaluation workflows, using SHAP to interpret model behavior and translate complex results into actionable research insights.
- Applied PCA, clustering, and graph analytics to examine patterns across 16 industries and four global regions and investigate operational and supply-chain risks across 30 global IT firms over a 10-year period.

RAG AI Agent & MLOps

- Developed a Python-based retrieval-augmented generation (RAG) application using NLP, text embeddings, vector search, and LLMs to retrieve relevant information from unstructured documents and generate context-grounded responses.
- Implemented retrieval ranking, response validation, and REST API integration to support information retrieval and AI-assisted decision-making for non-technical users.

Real-Time Data Engineering Platform

- Built a distributed data pipeline using PostgreSQL, Kafka, and Debezium for change data capture (CDC), with Airflow orchestration and Snowflake/dbt transformations to support near-real-time analytical workflows.
- Implemented Docker-based services, GitHub Actions CI/CD, and automated data validation to streamline pipeline testing, deployment, and data-quality checks.

BioBERT Drug Sentiment Classification Model

- Fine-tuned a BioBERT transformer model on 2,900+ clinical text records to classify drug-related sentiment using domain-specific NLP techniques.
- Evaluated predictions on a held-out test set of 600+ records, using classification metrics, confidence analysis, and error inspection to assess model reliability.

CERTIFICATION

AWS Certified Machine Learning Engineer - Associate.
Python for Data Science, AI & Development - IBM.
Microsoft Certified: Azure AI Fundamentals.

EDUCATION

Master's in Data Science, University of Massachusetts, Dartmouth, MA **GPA: 3.90/4.00**